

HOMEWORK ASSIGNMENT 5 – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

QUESTION 1: Commute Times.

The United States Census Bureau routinely publishes data (including predictive data) on their website. According to one of their published sources¹, the 2024 average (mean) commute time among all U.S. residents ages 16 or over (specifically, among those who did not work from home) was 27.2 minutes.

Suppose Eric wants to determine whether or not the average commute time among San Luis Obispo residents in 2024 is less than the national average (as reported by the U.S. Census Bureau). To that end, he obtained a simple random sample of 100 San Luis Obispo commuters (back in 2024); shown below are some summary statistics of Eric's data:

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 26.5$ min.	median = 23.0 min.	$s = 4.805$ min.	$n = 59$

Table 1: Summary statistics of Eric's data.

- (a) Describe one observational unit, in the context of this study.

Solution: One observational unit is **one person** (specifically, **one San Luis Obispo resident**).

- (b) Define the parameter in words, and identify its appropriate mathematical symbol.

Solution: The parameter of interest is the **average commute time among all San Luis Obispo residents**, which we denote by μ .

- (c) Write down the null and alternative hypotheses using symbols and math only.

Solution: Letting μ be defined as in the previous part, our hypotheses are

$$\begin{cases} H_0 : \mu = 27.2 \\ H_A : \mu < 27.2 \end{cases}$$

Note that we knew to use a lower-tailed alternative by the way the research conjecture is phrased: Eric wants to determine whether or not the average commute time among San Luis Obispo residents in 2024 is **less than** the national average.

- (d) Can we use a theory-based approach to test these hypotheses? Why or why not? **Hint:** is there any information missing?

Solution: The answer to this question is - **we don't know!** Or, if you wanted something more definitive, "**no, not with the information that is provided.**" In particular, remember that there are TWO parts to the validity conditions for quantitative data: not only do we need a large sample size (which we do have here - 59 is greater than or equal to 20), but we also need **a sample distribution that is not too heavily skewed**. We were not provided with the sample distribution, meaning we cannot determine whether a theory-based approach is appropriate or not.

¹ <https://www2.census.gov/programs-surveys/commuting/guidance/acs-1yr/Mean-travel-time.pdf>

Use the following for the remaining parts of this problem: We will proceed using a simulation-based approach. The plot below shows a dotplot representing a simulated null distribution, where I used **150** simulated samples (note that I did not run our usual 10,000 repetitions!)

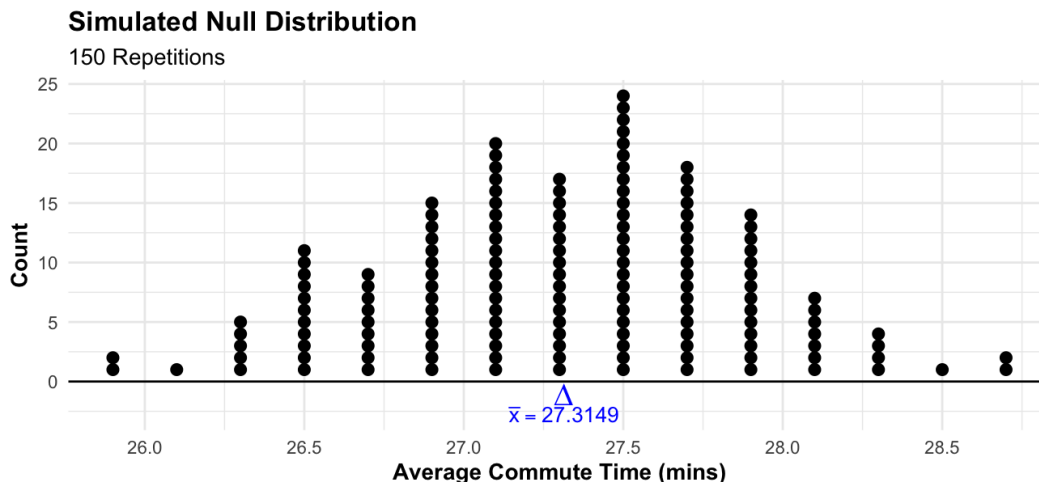


Figure 1: Simulated null distribution.

The mean of this simulated null distribution is 27.32 and the standard deviation (also of this simulated null distribution) is 0.55.

- (e) Calculate and report the p -value. Show your work!

Solution: The p -value in a simulation-based approach is the proportion of sample statistic values (in the simulated null distribution) that are as or more extreme as what was observed. We observed $\bar{x} = 26.5$, meaning we need to calculate the proportion of dots on Figure (1) that are less than or equal to 26.5. This gives us an approximate p -value of around

$$\frac{11 + 5 + 1 + 2}{150} = \frac{19}{150} \approx \mathbf{0.1267}$$

A Grading Note: don't forget to include the "as" part of "as or more extreme-" that is, don't forget to include the statistic values in the simulated null distribution that are equal to the observed statistic value.

- (f) Calculate the simulation-based t -statistic.

Solution: $t = \frac{\bar{x} - (\text{simulated mean of null distribution})}{(\text{simulated SD of null distribution})} = \frac{26.5 - 27.32}{0.55} \approx \mathbf{-1.490}$

- (g) Calculate the theory-based t -statistic (regardless of whether you think a theory-based approach is valid here or not).

Solution: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{26.5 - 27.2}{4.805/\sqrt{59}} \approx \mathbf{-1.119}$

- (h) How justified are we in generalizing the results of this study to the population of interest? In other words, describe the potential scope of inference of this study based on the sampling method described.

Solution: Because Eric's sample is a simple random sample, we likely have a **somewhat broad scope of inference**, since simple random sampling tends to be an unbiased sampling method. Stated differently, we do not have any reason to believe that the 59 SLO residents included in Eric's data differ from a "typical" SLO resident in terms of their commute times.

Note: It may be tempting to argue that SLO is a small city, and therefore the commute time of a typical SLO resident is not representative of the commute time of a typical US resident. However, note the research question: Eric is not trying to assess whether the average commute time of *all US residents* is less than 27.2 minutes or not - rather, he's interested in trying to assess whether the average commute time of *all SLO residents* is less than 27.2 minutes or not. This is why it's important to keep in mind the population of interest.

- (i) Answer the research question. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p -value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.

Solution: Based on the observed sample mean of $\bar{x} = 26.5$ minutes (statistic) we obtained a large p -value of 0.1267, which constitutes weak evidence against the null hypothesis, so we fail to reject the null hypothesis. Therefore, we do NOT have sufficient evidence to claim that the population mean commute time among all San Luis Obispo residents is less than the nationally-reported average of 27.2 minutes.

Some Grading Notes: As we get further into this course, I'll become more and more particular about how you phrase your conclusions. To help you understand my expectations, here are some comments I would have used if I were grading your submissions:

- I would have deducted points if you did not clearly indicate that the conclusion is a statement about a POPULATION mean.
- I would have deducted points if you did not clearly indicate WHICH POPULATION the parameter is referring to (in this case, the set of all SLO residents).
- I would have deducted points if you said "accept" the null. Again, we either REJECT or FAIL TO REJECT.

Of course, there are a few other things I'd look for (making sure you reported the correct statistic, making sure you're properly phrasing the null and/or alternative, etc.) but these are some highlights I wanted to share with at this stage.

- (j) Eric's friend, Erika, is interested not in whether the average commute time among San Luis Obispo residents is less than the national average (of 27.2 minutes), but merely whether the average commute time among San Luis Obispo residents *differs* from the national average. If Erika uses the same data that Eric did, what would be her approximate p -value? Do **not** use any applets for this; also, as a hint, you shouldn't need to re-count anything (though you may need to recalculate certain quantities).

Solution: Because Erika’s test is a two-sided test and Eric’s original p -value is less than 0.5, Erika’s p -value should be about twice that of Eric’s: $2 \cdot 0.1267 \approx \mathbf{0.2534}$.

Alternatively, if we really wanted to, we could have recounted dots on the dotplot:

$$\frac{19}{150} + \frac{7 + 4 + 1 + 2}{150} = \frac{33}{150} \approx 0.22$$

QUESTION 2: Reaching New Heights.

Based on statistics from the [Centers for Disease Control](#) (CDC), we can infer that they claim the average height of a US resident is 66.2 inches.² Suppose I (Dr. Ethan) think that’s a little low. In particular, I’d like to address the question: is the true average height of a US resident larger than 66.2 inches?

As a part of the Welcome Survey I sent out at the start of the semester, I asked all of you (all three of my STAT 1110 lecture sections) what your height is in inches, without shoes. After cleaning up responses (converting feet to inches, removing obvious mis-inputs, etc.) I was left with a total of 82 responses. Shown below is a dotplot of these 82 height measurements, along with some summary statistics.

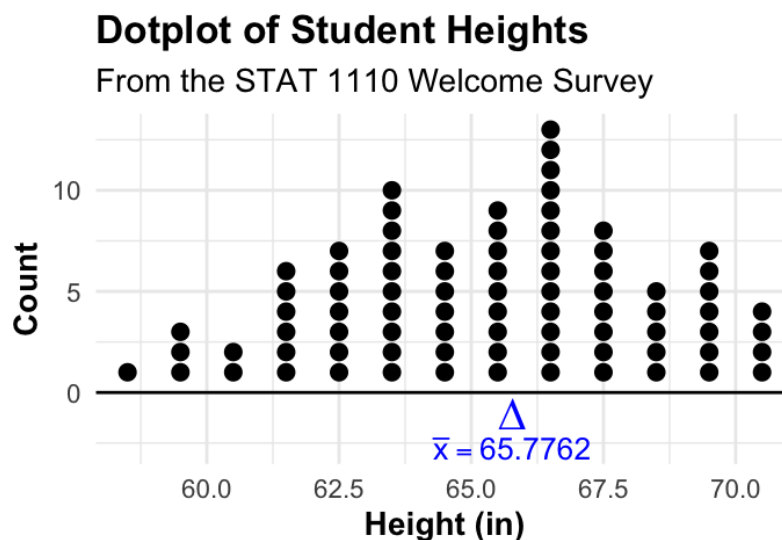


Figure 2: Results from the Welcome Survey.

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 65.776$ in.	median = 66 in.	$s = 2.999$ in.	$n = 82$

Table 2: Summary statistics of the data from the Welcome Survey.

(a) What does one dot on Figure (2) represent?

²Technically they only report an average height for “men” and “women.”

Solution: One dot represents **one student** (or, **one student and their height**).

- (b) Define the parameter of interest, and identify its appropriate mathematical symbol.

Solution: The parameter of interest is the **average height of all US residents**. Since this is a population mean, we use the symbol μ to denote it.

- (c) Write down the null and alternative hypotheses using proper mathematical symbols and numbers.

Solution:

$$\begin{cases} H_0 : \mu = 66.2 \\ H_A : \mu > 66.2 \end{cases}$$

- (d) Are the validity conditions met if we want to conduct an analysis based on the theory-based approach (i.e. a one-sample t -test)? Explain.

Solution: Yes. There are two things we need to note:

- The sample size ($n = 82$) is greater than or equal to 20
- The sample distribution (Shown in Figure 2) is not too heavily skewed

Regardless of your answer to the previous part, we'll proceed with a theory-based approach for the remainder of this problem.

- (e) Calculate the value of the t -statistic.

Solution: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{65.776 - 66.2}{2.999/\sqrt{82}} \approx -1.28$

- (f) Navigate to the [Theory-Based Inference Applet](#) and calculate the theory-based p -value.

Solution:

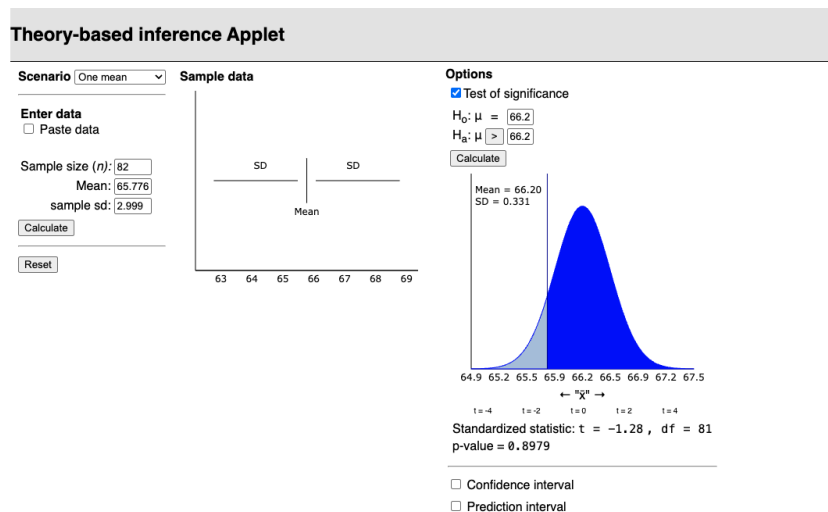


Figure 3: A screenshot from the Theory-Based Inference Applet.

The p -value is around **0.8979**.

- (g) **Evaluate** and **Interpret** the p -value. (At this point, I will rely on you to remember what these mean!)

Solution: When **evaluating** a p -value, we are assessing the strength of evidence afforded against the null. Because a p -value of 0.8979 is quite large, the data affords **Not much evidence against the null hypothesis**.

When **interpreting** a p -value, we need to essentially rephrase the definition of a p -value in the context of the study. Here's how I would do that:

The p -value of 0.8979 is the probability of observing a sample mean height (in inches, of a sample of 82 people) less than or equal to 65.776 inches, assuming the true average height of all US residents is 66.2 inches.

- (h) How justified are we in generalizing the results of this study to the population of interest? In other words, describe the potential scope of inference of this study based on the sampling method described.

Solution: This is a little open-ended. But, it's important to note that the data is essentially a **convenience sample**, and convenience sampling has a potential to be biased. However, most students in STAT 1110 are probably done growing, meaning as far as height is concerned STAT 1110 students may still be representative of the population (US residents). For example, if we had taken a sample of kindergarten students, we'd think that our sampling method may underestimate the true value of the parameter of interest.

- (i) Answer the research question. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p -value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) **reject/fail to reject the null hypothesis**, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.

Solution: Based on the observed sample mean of $\bar{x} = 65.776$ in, we obtained a large p -value of 0.8979, which constitutes very weak evidence against the null hypothesis. As such, we fail to reject the null hypothesis. That is, we do **NOT** have sufficient evidence to claim that the true population mean height across all US residents is greater than 66.2 inches.

Some Grading Notes: Again, as we get further into this course, I'll become more and more particular about how you phrase your conclusions. To help you understand my expectations, here are some comments I would have used if I were grading your submissions:

- I would have deducted points if you did not clearly indicate that the conclusion is a statement about a POPULATION mean.
- I would have deducted points if you did not clearly indicate WHICH POPULATION the parameter is referring to (in this case, the set of all US residents).
- I would have deducted points if you said "accept" the null. Again, we either REJECT or FAIL TO REJECT.

Of course, there are a few other things I'd look for (making sure you reported the correct statistic, making sure you're properly phrasing the null and/or alternative, etc.) but these are some highlights I wanted to share with at this stage.

QUESTION 3: Notation Check-In

In each of the following situations, you will be presented with several quantities. Identify the correct mathematical symbol for the indicated quantity.

- (a) Pia wants to determine whether the the proportion of households in a rural county that live under the poverty line exceeds the national average. She took a sample of 12 households, and found that 8 of them fall under the poverty line. **What symbol should be used to denote the quantity 8/12?**

Solution: This is a **sample proportion**, and is therefore denoted by $\hat{p}$.

- (b) Katrina wants to determine the average weight across all hummingbirds in California. A convenience sample of 50 hummingbirds had an average weight of 5.34 grams.

- i. What symbol should be used to denote the quantity 5.34 grams?

Solution: This is a **sample mean**, and is therefore denoted by $\bar{x}$.

- ii. What symbol should be used to denote the quantity 50?

Solution: This is a **sample size**, and is therefore denoted by n .

- (c) Sneha has read an article that claims the average height of all palm trees in the Brazilian Amazon is 57.41 feet. She takes random sample of 64 palm trees, and finds this sample of trees to have an average height of 54.32 feet and a standard deviation of 12.43 feet. Unbeknownst to Sneha or the authors of the article, the set of all palm trees in the Brazilian Amazon have an average height of 59.23 feet and a standard deviation of 17.44 feet.

- i. What symbol should be used to denote the quantity 57.41 feet?

Solution: This is the true value of the parameter under the null - more specifically, the **value of a population mean, under the null**. Therefore, we use the symbol μ_0 to denote it.

- ii. What symbol should be used to denote the quantity 54.32 feet?

Solution: This is a **sample mean**, and is therefore denoted by $\bar{x}$.

- iii. What symbol should be used to denote the quantity 12.43 feet?

Solution: This is a **sample standard deviation**, and is therefore denoted by s .

- iv. What symbol should be used to denote the quantity 59.23 feet?

Solution: This is a **population mean**, and is therefore denoted by μ . **Note:** as has been stated many times before, we do not typically know the true value of a parameter in practice!

- v. What symbol should be used to denote the quantity 17.44 feet?

Solution: This is a **population standard deviation**, and is therefore denoted by σ .

- vi. What symbol should be used to denote the quantity 64 feet?

Solution: This is a **sample size**, and is therefore denoted by n .

QUESTION 4: Type I and Type II Errors

As with a jury trial, another analogy to hypothesis testing involves medical diagnostic tests. These tests aim to indicate whether or not the patient has a particular disease. But the tests are not infallible, so errors can be made. The null hypothesis can be regarded as the patient being healthy (the baseline). The alternative hypothesis can be regarded as the patient having the disease.

(a) Describe what Type I error (false alarm) represents in this situation.

- ✓ **The patient does not have the disease, but the test says the patient has the disease**
- The patient has the disease, but the test says the patient is healthy.

Solution: A Type I error happens when we reject the null, even though the null was true. Saying “the null was true” in the context above is equivalent to saying “the person was healthy (does not have the disease),” and saying “we reject the null” is equivalent to saying “the test rejected the claim that the person was healthy (i.e. it said the person has the disease).”

(b) Describe what Type II error (missed opportunity) represents in this situation.

- The patient does not have the disease, but the test says the patient has the disease
- ✓ **The patient has the disease, but the test says the patient is healthy.**

Solution: A Type II error happens when we fail to reject the null, even though the null was false. Saying “the null was false” in the context above is equivalent to saying “the person was not healthy (has the disease),” and saying “we fail to reject the null” is equivalent to saying “the test failed to reject the claim that the person was healthy (i.e. it said the person is healthy).”